PROJECT DYSON SPHERE

How to Read Every Simulation Plot

A source-grounded visual textbook for HelioSwarm M1–M6

Axes → legend → physical interpretation → success tests → objectives O1–O5

Based on branch `helioswarm-m0-m6`, commit `560675f`

13 November 2026 | Companion to *Mathematical Foundations M1–M6*

Before you read: an evidence and figure guide

Key interpretation

There are two distinct criteria for success. A *computationally successful* plot shows that the code solves and checks the stated mathematical model. An *engineering-successful* plot shows that the resulting spacecraft architecture can deliver useful power under credible physical, mass, safety, environmental and operational constraints. A negative feasibility result can be computationally excellent.

What is actually in GitHub? At the inspected commit, the branch commits exactly **five PNG figure files**: two M6 single-satellite composite plates and three M6b characterization plates. M1 and M2 have numerical validation results in text, but no committed standalone figures. The executable plotter also writes `orbit_access.png`, `solar_thermal.png`, `beam_capture.png`, `power_stages.png` and `unavailability.png` inside run directories; these individual outputs were not committed as separate PNGs in the inspected branch. The present textbook covers *every panel of the committed PNGs*, adds explicitly labelled diagnostic reconstructions for M1 and M2, and replots the two committed sweep CSV tables.

Image provenance matters. The graphs embedded here are *new, readable teaching redrawings*, not pixel-for-pixel copies of the five GitHub PNGs. Some are direct evaluations of published model equations and configuration parameters, some replot *recorded CSV measurements*, and others use *schematic or calibrated traces*. Their status is stated under each illustration. **Never cite an illustrative curve as a new numerical simulation result.** Each illustration is clickable and points to its relevant original figure or source documentation on GitHub; the exact original five PNGs are catalogued in the source index. This division is necessary because the original PNG bytes were accessible for inspection but not for local PDF embedding from the private repository connector.

Document	Scope	Evidence status
`mvp_optical_composite.png`	O1/orbit, O2/solar, O3/optical beam, M6/grid losses; 11 live panels plus notes	Original PNG linked
`mvp_microwave_composite.png`	Sail panel arrangement, microwave technology assumptions	Original PNG linked
`power_thermal_vs_distance.png`	Six curves versus heliocentric radius	Original PNG linked
`capture_vs_pointing.png`	Two receiver-capture/pointing panels	Original PNG linked
`capture_vs_wavefront.png`	Wavefront phase-error response, two y axes	Original PNG linked

How to read the next chapters. For every panel, identify its x and y quantities and their *units*, distinguish measured/derived/schematic series, understand why the curve changes, identify the validation criterion, and ask whether the physical finding answers an O-level research question. *M* means software milestone, *O* scientific objective, *E* recorded experiment, and *L* orbital model level.

Contents

Before you read: evidence and figure guide i

Plot-reading grammar and vocabulary iv

1 M1 – Analytic Keplerian dynamics and what a validated orbit plot means 1

1.1 Panel M1-A: Bound two-body trajectories . . . 1

1.2 Panel M1-B: Mean anomaly versus true anomaly . . . 1

1.3 Panel M1-C: Two-body conservation residual . . . 2

2 M2 – Planetary perturbations, solar radiation pressure and validation diagnostics 3

2.1 Panel M2-A: Trajectory displacement relative to the L0 reference . . . 3

2.2 Panel M2-B: Numerical tolerance convergence . . . 3

2.3 Panel M2-C: Sensitivity to collector area-to-mass ratio . . . 4

3 M3 – Targeting the moving Earth and knowing when a link is possible 5

3.1 Panel M3-A: Satellite orbit – J2000 / SSB projection . . . 5

3.2 Panel M3-B: Earth-receiver range and heliocentric distance . . . 5

3.3 Panel M3-C: Elevation and access fraction . . . 6

3.4 Panel M3-D: Daily elevation windows – zoomed view . . . 6

3.5 Panel M3-E: Vacuum light-time . . . 7

3.6 Panel M3-F / M6: Unavailability categories . . . 7

3.7 Additional M3 geometric quantities not plotted separately . . . 7

4 M4 – Solar collection and the thermal boundary of feasibility 8

4.1 Panel M4-A: Solar collection over one year . . . 8

4.2 Panel M4-B: PV-temperature threshold over time . . . 8

4.3 Panel M4-C: Solar flux versus heliocentric distance . . . 9

4.4 Panel M4-D: Collector electrical power . . . 9

4.5 Panel M4-E: Temperature-dependent PV conversion efficiency . . . 9

4.6 Panel M4-F: PV thermal limit and red infeasible marks . . . 10

4.7 Panel M4-G: Radiator waste heat . . . 10

4.8 Panel M4-H: Required radiator area . . . 10

5 M5 – Diffraction, finite-receiver capture, pointing and wavefront errors 11

5.1 Panel M5-A: Airy intensity profile of a circular transmitter . . . 11

5.2 Panel M5-B: Encircled energy / finite receiver capture . . . 11

5.3 Panel M5-C: Inverse scaling with aperture diameter . . . 12

5.4 Panel M5-D: Optical beam capture over time . . . 12

5.5 Panel M5-E: Microwave beam capture over time . . . 13

5.6 Panel M5-F: Pointing efficiency versus time . . . 13
5.7 Panel M5-G: Fresnel number . . . 13
5.8 Panel M5-H: Absolute capture versus pointing jitter . . . 14
5.9 Panel M5-I: Relative pointing efficiency versus jitter . . . 14
5.10 Panel M5-J: Expected capture versus RMS wavefront error . . . 15
5.11 How to interpret atmospheric attenuation across M5 panels . . . 15

6 M6 – Reading the complete link budget, annual grid power and sweep outputs 17
6.1 Panel M6-A: The optical power-stage "waterfall" . . . 17
6.2 Panel M6-B: Microwave power-stage waterfall . . . 18
6.3 Panel M6-C: Optical grid-power time series . . . 18
6.4 Panel M6-D: Microwave grid-power time series . . . 18
6.5 Panel M6-E: Numerical summary text box in both composites . . . 19
6.6 Panel M6-F: Capacity factor in the grid-power plot title . . . 19
6.7 Panel M6-G: The composite's idealized-model notes . . . 19

7 M6 parameter sweeps – a bridge toward O4, not constellation optimization 20
7.1 Panel M6-H: Optical mean grid-power heatmaps . . . 20
7.2 Panel M6-I: Microwave sweep heatmap . . . 21
7.3 Panel M6-J: Microwave size-sensitivity line plot . . . 21
7.4 What O4 needs beyond these sweeps . . . 21

8 Crosswalk: which plots answer which research objective? 22
8.1 Do not confuse the three meanings of "successful" . . . 22
8.2 Practical checklist for a manuscript figure . . . 22

A Exact original plot and report catalogue 24

B Terminology quick-reference 25

Final synthesis 26

Plot-reading grammar and technical vocabulary

Time series: a quantity plotted against time, usually days since 2030-01-01 in the MVP. **Abscissa** is the horizontal x axis and **ordinate** the vertical y axis. **Linear scale** gives equal spacing for equal changes; a **logarithmic scale** gives equal spacing for equal ratios. Thus 10^{-3} to 10^{-2} is a tenfold rise, not a 0.009-unit visual increment. **Dimensionless fraction** ranges from 0 to 1; multiply by 100 to obtain percent. **Percentile p95** is a value with 95% of samples at or below it, not a 95% confidence claim. **Shaded percentile band p05–p95** describes spread under an assumed distribution, not necessarily uncertainty in the physical model.

Availability is the fraction of simulated time when all hard link conditions allow operation. It differs from **capture efficiency**, the fraction of an emitted beam intercepted by a finite receiver, and from **end-to-end efficiency**, the ratio of grid energy to sunlight incident on the collector. **Capacity factor** divides mean grid power by a particular *reference-link rating*; it is not PV efficiency. **Outage** is a maximal contiguous unavailable interval. **Steady state** treats heat production and rejection as instantaneously balanced. **Thermally feasible** means the modeled PV and radiator temperatures satisfy specified limits; it does not imply manufacturable radiator mass.

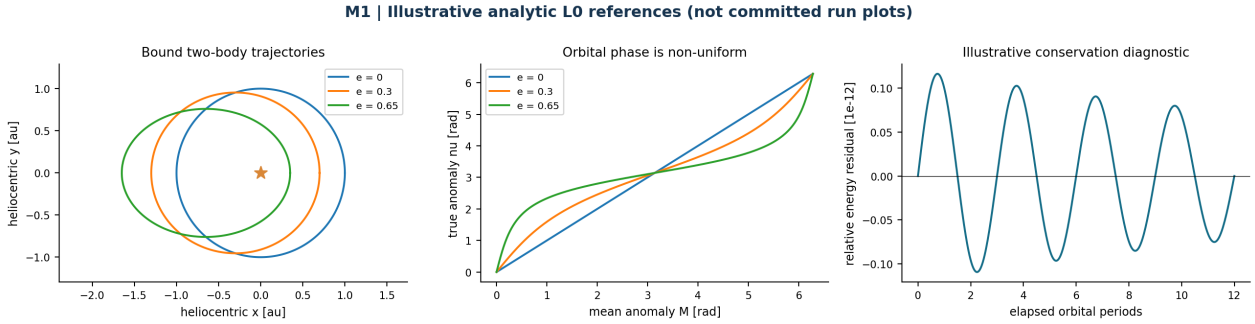
au (astronomical unit) = 149,597,870,700 m. **W** is watt, instantaneous power (J/s); **Wh** is watt-hour, energy (3600 J); 1 MW = 10^6 W; 1 MWh = 3.6×10^9 J. **Kelvin** is absolute temperature and has the same increment size as Celsius. **Radian** is angular measure; 1 rad $\simeq 57.296^\circ$. **Heliocentric** means Sun-centered. **SSB** is the Solar System Barycenter. **J2000** is a fixed inertial coordinate orientation. **PV** means photovoltaic conversion. **RF** means radio-frequency electromagnetic radiation. **Rectenna** is a rectifying antenna that converts RF into DC electricity. **Diffraction** is wave spreading set by wavelength and finite transmitter aperture.

M1 – Analytic Keplerian dynamics and what a validated orbit plot means

Key interpretation

Connection: M1 → O1. M1 supplies the exact two-body baseline against which later, more realistic trajectories must be compared. M1 does *not* decide where the constellation should orbit; it validates the basic orbital coordinate and propagation machinery.

Figure status: M1 has no standalone PNG committed at this branch. The following three panels are pedagogical analytic checks, not extracted M1 run figures.



1.1 Panel M1-A: Bound two-body trajectories

Horizontal axis: x position relative to the Sun in astronomical units (au) **Vertical axis:** y position relative to the Sun in au

Curves. Each closed curve has semimajor axis $a = 1$ au but a different eccentricity $e \in \{0, 0.3, 0.65\}$. The orange star is the focus containing the Sun, not necessarily the ellipse's center. **Eccentricity** e is the dimensionless departure from circularity: $e = 0$ is a circle and $0 < e < 1$ is a closed ellipse. **Physical meaning.** Increasing e changes perihelion $r_{\text{peri}} = a(1-e)$ and aphelion $r_{\text{aph}} = a(1+e)$ while retaining the same period for fixed a and solar gravitational parameter. These positions are calculated from $r = a(1-e^2)/(1+e\cos\nu)$, where ν is true anomaly. Apparent orbital shape depends on coordinate projection and axis aspect ratio.

What to check. Do not accept a visually circular line as a stability test. Compute state-element-state round trips, check that the trajectory closes, and verify the appropriate two-body invariants.

What a successful result looks like

A circular case must maintain constant r to numerical precision. Each elliptical case must pass through $a(1-e)$ and $a(1+e)$ and return to its initial state after $T = 2\pi\sqrt{a^3/(GM_\odot)}$, within the code's tested floating-point tolerances.

Research objective. This establishes O1's geometric and propagation reference, but does not yet establish real heliocentric stability under other bodies or radiation forces.

1.2 Panel M1-B: Mean anomaly versus true anomaly

Horizontal axis: mean anomaly M [radians] **Vertical axis:** true anomaly ν [radians]

Mean anomaly increases uniformly with time in the Kepler problem: $M(t) = M_0 + n(t - t_0)$ and $n = \sqrt{GM_\odot/a^3}$ is mean motion. **True anomaly** is the physical polar angle measured from perihelion to the spacecraft's position. **Eccentric anomaly** E is a convenient auxiliary angle satisfying $M = E - e \sin E$. **Physical meaning.** The $e = 0$ line is essentially $\nu = M$. Higher-eccentricity lines bend because a spacecraft moves faster near perihelion and slower near aphelion (Kepler's second law). The mismatch of ν and M is expected physical behavior, not a bug.

What to check. Kepler's equation must have a unique solution for $0 \leqslant e < 1$ and the residual $|E - e \sin E - M|$ must be small after angle reduction. A curve with discontinuous jumps other than chosen 2π wrapping would need explanation.

1.3 Panel M1-C: Two-body conservation residual

Horizontal axis: elapsed number of orbital periods **Vertical axis:** relative mechanical-energy residual, shown in units of 10^{-12}

Residual means the difference between a computed conserved quantity and its initial value. The specific orbital energy is $\varepsilon = v^2/2 - GM_\odot/r$ (J/kg), while specific angular momentum is $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ (m^2/s). In an ideal isolated two-body orbit, both remain constant. **Physical meaning.** A bounded, near-zero residual indicates that rounding rather than secular numerical drift governs the analytic solution. The blue wiggle here is schematic; it is not a measured repository residual.

What a successful result looks like

The success criterion is a quantitative tolerance on $\max_t |\varepsilon(t) - \varepsilon_0|/|\varepsilon_0|$ and $|\mathbf{h}(t) - \mathbf{h}_0|/|\mathbf{h}_0|$, not aesthetic smoothness. M1 documents explicit high-precision tests and angle-singularity conventions for circular and equatorial orbits.

Assumptions / limitations

M1 uses a massless spacecraft, point-mass Sun and no planetary perturbations, solar radiation pressure or relativity. Accordingly an M1 plot cannot establish the actual orbit of a physically extended, reflecting collector.

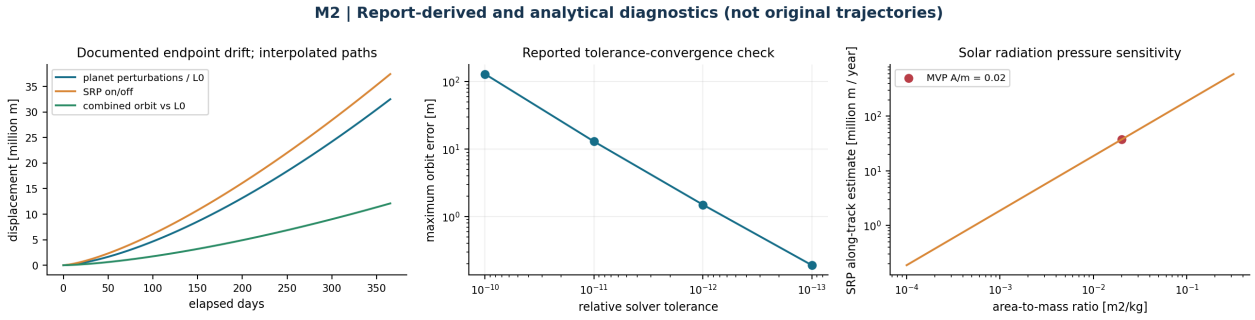
PLOT INTERPRETATION 2

M2 – Planetary perturbations, solar radiation pressure and validation diagnostics

Key interpretation

Connection: M2 → O1. M2 introduces the L1 trajectory used by later plots and tests its fidelity against the L0 reference and the independent REBOUND N-body integrator. The relevant achievement is correct force modeling and numerical convergence, *not* a plot that stays identical to Kepler.

Figure status: No standalone M2 PNG is committed. The following page redraws diagnostics from documented endpoint numbers and analytic SRP formulas. Only the convergence points are directly transcribed results; time-path shapes are illustrative interpolations.



2.1 Panel M2-A: Trajectory displacement relative to the L0 reference

Horizontal axis: elapsed time [days] **Vertical axis**: position difference between two simulated models [million metres]

Drift is defined by $\Delta r(t) = \|\mathbf{r}_A(t) - \mathbf{r}_B(t)\|$, measured between trajectories at the same epoch and in the same frame and origin. **Planetary perturbations** are accelerations from the Solar System's planets and Moon. **Solar radiation pressure (SRP)** is acceleration caused by solar photons transferring momentum to the collector. **Physical meaning.** The end-of-year reported differences are approximately 3.25×10^7 m from planets alone, 3.742×10^7 m for SRP-on versus SRP-off and 1.21×10^7 m for the combined model versus L0. The smaller combined difference is not contradictory: perturbation vectors can partly cancel. The curved histories drawn here interpolate these endpoints and are *not* source-computed trajectories.

What to check. All comparisons must use the same initial state, identical reference frames and epoch sampling, and an explicit statement of which forces change. A model difference is not automatically an integration error.

2.2 Panel M2-B: Numerical tolerance convergence

Horizontal axis: DOP853 relative error tolerance, logarithmic **Vertical axis**: trajectory comparison error [m], logarithmic

DOP853 is an adaptive, eighth-order Runge–Kutta integrator. **Relative tolerance** is the requested proportional numerical error control; reducing it typically increases computational cost and accuracy until rounding or model error dominates. **Physical meaning.** The reported test values decrease approximately 130 m → 13 m → 1.5 m → 0.19 m as relative tolerance tightens from 10^{-10} to 10^{-13} for a particular Sun-only comparison. Such a systematic decrease supports numerical convergence.

What a successful result looks like

The solution should become insensitive to tighter tolerances, smaller maximum integration steps and finer ephemeris interpolation, within a predefined error budget. Importantly, the L1 full-body solution agreed with REBOUND to approximately 1.03 m over the tested year; that is an independent gravitational-model check.

2.3 Panel M2-C: Sensitivity to collector area-to-mass ratio

Horizontal axis: solar-illuminated cross-sectional area divided by spacecraft mass, A/m [m^2/kg], logarithmic **Vertical axis:** estimated SRP-induced along-track drift after one year [million m], logarithmic

Area-to-mass ratio controls the acceleration imparted by a given radiation pressure. **Reflectivity coefficient** C_r parameterizes absorbed/reflected momentum transfer. The source model uses $P_{\mathrm{SRP}} = S(r)/c$ and $a_{\mathrm{SRP}} = P_{\mathrm{SRP}} C_r A/m$ for a radial Sun-facing approximation. **Physical meaning.** The linear trend on log–log axes follows the small-SRP estimate $\Delta s \simeq 2\beta nta$ with $\beta \propto C_r A/m$. The MVP assumes $A/m = 0.02$ m^2/kg and $C_r = 1.3$, producing order 3.74×10^7 m per year in its SRP-on/off comparison.

Assumptions / limitations

The L1 radial-SRP model does not represent changing collector incidence, structural attitude control, non-radial reflection or solar-sail steering. The model level L3 would refine these assumptions. Strong SRP response does not prove orbit instability; it says the chosen force must be included.

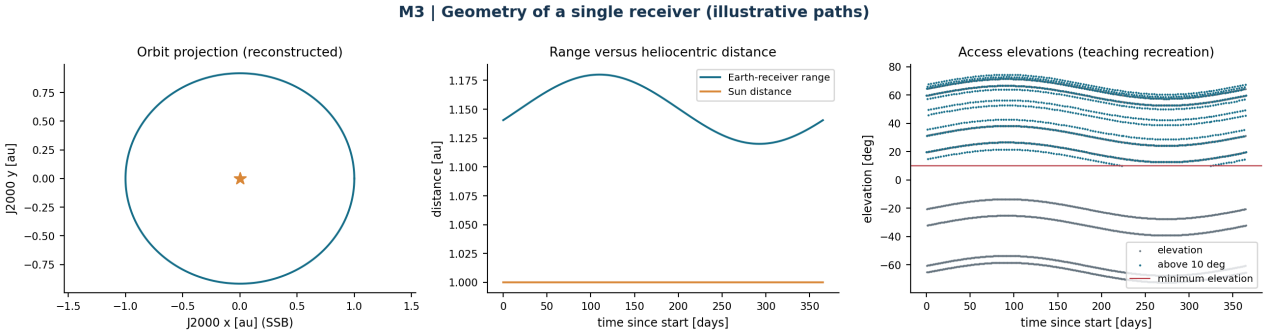
Research objective. O1 is substantially supported for one satellite by the validated dynamic propagator. The figure cannot answer O1's eventual constellation-phase and long-term station-keeping optimization questions.

M3 – Targeting the moving Earth and knowing when a link is possible

Key interpretation

Connection: M3 → O1 and O3. M3 turns a valid heliocentric orbit into the target range, arrival time, elevation and line-of-sight constraints required by a physical power link. It directly answers the *single-satellite access* part of O1 and provides geometry to O3.

Figure status: The committed optical and microwave composites each contain the three M3 orbit/access panels and the M6 unavailability panel. Below is a source-shaped teaching recreation. Exact original: optical composite.



3.1 Panel M3-A: Satellite orbit – J2000 / SSB projection

Horizontal axis: J2000 x coordinate [au] **Vertical axis**: J2000 y coordinate [au]

J2000 supplies a fixed nonrotating orientation; **SSB** is the center of mass of the Solar System, not the Sun's exact position. The original plot marks the SSB with an orange star, while the trajectory is the x–y projection of the spacecraft's three-dimensional position. Its near-elliptical appearance is affected by J2000's tilt relative to the ecliptic plane. The plotted coordinates are an *inertial position projection*, not a drawing of beam shape or energy flow. **Physical meaning.** A smooth closed-looking track is consistent with a nearly circular one-year heliocentric orbit. It is not enough to claim dynamical stability: perturbations may accumulate on longer timescales without visibly affecting the one-year orbit picture. The Sun and Earth should not be inferred to share the plotted origin.

What to check. Check the plotted values against the saved `sat_x_m` and `sat_y_m` columns after division by 1 au; confirm the same frame and origin were used for both coordinates.

What a successful result looks like

A successful M3 projection faithfully represents the M2 ephemeris trajectory; scientific O1 success additionally requires distances, visibility and perturbation sensitivity. A graph can look smooth and still use a wrong coordinate origin.

3.2 Panel M3-B: Earth-receiver range and heliocentric distance

Horizontal axis: time since the 2030-01-01 start [days] **Vertical axis**: distance [au]

Blue (Earth-receiver range) is $R(t) = \|\mathbf{r}_{\text{receiver}}(t_a) - \mathbf{r}_{\text{sat}}(t_e)\|$, using emission time t_e and arrival time t_a. **Orange (Sun**

distance) is $r_{\odot}(t) = \|\mathbf{r}_{\mathrm{sat}}(t) - \mathbf{r}_{\odot}(t)\|$. The curves refer to distinct physical separations. **Physical meaning**. In the recorded optical baseline, $r_{\odot} \approx 1$ au remains almost flat while Earth-receiver range changes between about 1.122 and 1.180 au. Because wavelength and transmitter diameter are fixed, an increase in R broadens the diffraction footprint. The model reports light travel times from approximately 559.9 to 588.9 seconds: not negligible compared with motion of Earth over several minutes.

What to check. Never confuse the Sun-distance curve with the path length to Earth's ground receiver. A Sun-centered 1-au orbit does *not* imply the spacecraft is close to Earth.

3.3 Panel M3-C: Elevation and access fraction

Horizontal axis: time since start [days] **Vertical axis**: local receiver elevation [degrees]

Elevation e_l is the angle from the receiver's horizon to the apparent spacecraft direction. Values $e_l < 0^{\circ}$ are below the geometric horizon; 0° is the horizon and 90° is zenith. The **red horizontal line** is the configured 10-degree minimum elevation. The **gray series** shows the full elevation history; **blue points** denote accessible epochs, conditional on all geometric restrictions. **Physical meaning**. Dense bands result from daily visibility cycling across an entire year. Only 42.21% of the original hourly epochs are classified accessible. The blue areas are not a period of uninterrupted connectivity; access recurs in shorter windows. The graph is not a population of many satellites: multiple visible arcs correspond to repeated passes of the *same* spacecraft.

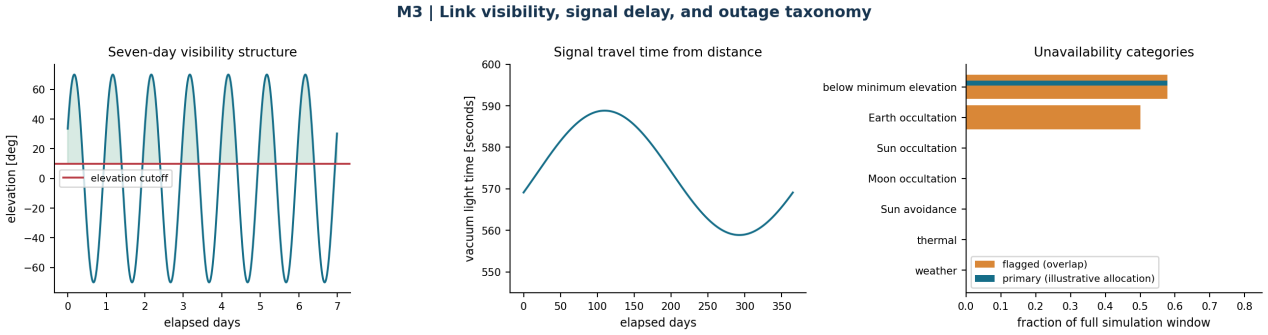
Assumptions / limitations

The original one-hour sampling interval suffers *diurnal aliasing*: nearly the same ground-hour angles are revisited each day. A follow-up run using a different, incommensurate time step found approximately 44.16% availability. Thus 42.21% is a discretized estimate, not a precision physical constant.

What a successful result looks like

A successful access model correctly handles Earth rotation, site geodesy, 10-degree elevation cutoff and beam occultation; successful analysis demonstrates time-step convergence. O1 still needs distributions across multiple orbital phases and receiver locations.

Figure status: The next three panels use source-reported ranges and reason categories but schematic elevation/light-time curves; they are not saved-run traces.



3.4 Panel M3-D: Daily elevation windows – zoomed view

Horizontal axis: elapsed time [days] **Vertical axis**: elevation angle [degrees]

In a zoomed graph, an individual day shows a smooth rise above the horizon, an interval in which $e_l \geqslant 10^{\circ}$, then a decline below the cutoff. The shaded region indicates the portion of the illustrative day when the cutoff is met. This is a teaching *zoom*; a real time series must be read from `timeseries.parquet`. **Access fraction** integrates the duration of these valid windows over the full simulated year, not their peak height. **What to check**. A passing result must not count an elevation-crossing point as an entire hour of availability without accounting for numerical time resolution. A robust result should report the longest access and outage intervals and their sampling uncertainty.

3.5 Panel M3-E: Vacuum light-time

Horizontal axis: elapsed days **Vertical axis**: propagation delay τ [seconds]

The source solves $c\tau = \|\mathbf{r}_{\text{site}}(t_e + \tau) - \mathbf{r}_{\text{sat}}(t_e)\|$ because the ground station moves before the photon arrives. $c = 299{,}792{,}458$ m/s is light speed in vacuum, and τ is light travel time. The displayed teaching curve is the simple distance-over-c relationship using a representative distance variation. **Physical meaning**. A delay of approximately 560–589 seconds means the beam must target Earth's *future* receiver position and account for aberration; aiming at the site's emission-epoch position is generally wrong. This is not a communication latency problem alone; it alters beam direction.

What a successful result looks like

The original M3 implementation uses a fixed-point light-time iteration with tolerance 10^{-9} s and a numerical fallback; the baseline converged in three iterations and reports a millimeter-scale path residual. A correct plot should agree with R/c in the static-receiver limit.

3.6 Panel M3-F / M6: Unavailability categories

Horizontal axis: reason labels on the vertical axis **Vertical axis**: time fraction [unitless, between 0 and 1]

This graph is *horizontal-bar oriented*: categories such as `BELOW_MIN_ELEVATION`, `EARTH_OCCULTATION`, Sun and Moon blockage, thermal failure, weather, pointing and safety abort are the labeled rows. A value of 0.50 means the stated flag is present approximately half the time. **Primary** causes assign *one* reason to each blocked epoch; **flagged** causes include simultaneous overlaps. Do not sum overlapping flagged fractions. **Physical meaning**. The original composite shows roughly 0.58 below-minimum-elevation and 0.50 Earth-occultation flagged. Their overlap is substantial; summing them would incorrectly predict over 100% blocked time. The primary bars sum to unavailable time, approximately $1 - 0.4221 = 0.5779$. No baseline outages are attributed to weather or thermal limits because the specified scenario assumes permanently clear sky and is thermally feasible at 1 au.

Assumptions / limitations

An absent weather bar under `weather_scenario=clear` is *not* a proof of all-weather optical transmission. Likewise, the lack of a Sun-avoidance bar is unsurprising when the minimum avoidance angle is set to zero. These are permissive baseline assumptions.

3.7 Additional M3 geometric quantities not plotted separately

Azimuth is the local compass angle from north through east toward the satellite; **Sun separation** is the angle between apparent satellite and Sun directions from the receiver; **occultation** means the beam segment intersects an opaque body; **umbra** denotes complete eclipse, **penumbra** partial eclipse. **Aberration** is the relativistic change of a photon propagation direction between moving frames; **point-ahead angle** is the difference between aiming at the receiver's current and future position; **pointing rate** is angular speed of the ideal beam direction (rad/s). These quantities are numerically computed and reported, even when not assigned a dedicated committed PNG. The source is `https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/docs/model_assumptions.md` (M3 section). **Research objective**. M3 answers how often a single satellite can see one receiver and how to target it. That supports O1 and provides the geometric inputs for O3; it cannot establish a continuous Dyson swarm without M7's independent satellite tracks and shared ground scheduling.

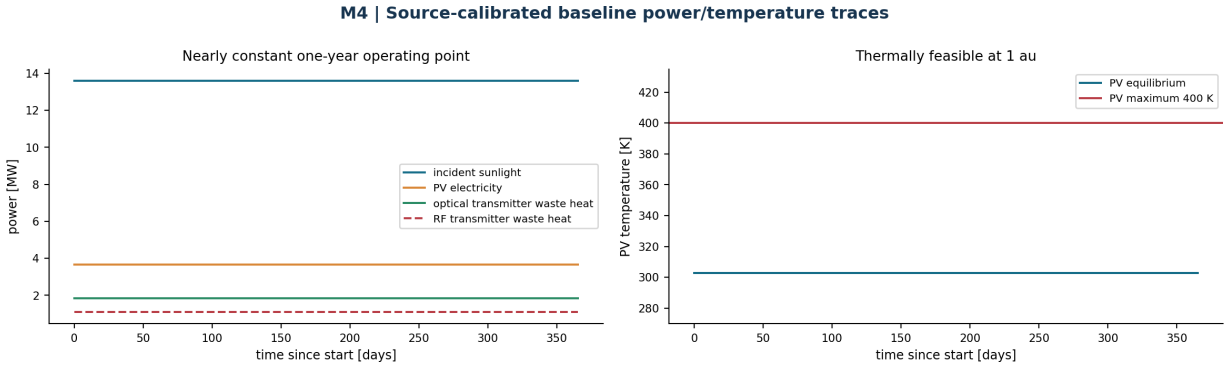
PLOT INTERPRETATION 4

M4 – Solar collection and the thermal boundary of feasibility

Key interpretation

Connection: M4 → O2. M4 must answer how much electricity a collector can produce at radius r and whether its cells and radiator stay below their prescribed temperatures. More solar irradiance does not guarantee a feasible spacecraft.

Figure status: The original composites each include a two-panel M4 section. The six-panel power/thermal characterization is separately committed. Redrawn below directly from the published equations and baseline configuration; this is a model reproduction, not a saved figure byte copy.



4.1 Panel M4-A: Solar collection over one year

Horizontal axis: time since start [days] **Vertical axis**: power [megawatts, MW]

The **incident sunlight** series is solar flux $S(r)$ multiplied by the projected collector area, before absorption or conversion. The **PV-electricity** series is the electrical output P_{PV} of the collector. The **electronics/transmitter waste-heat** series is the heat rejected by the independent radiator thermal node. Unlike the original composite, the teaching redraw overlays the *optical and microwave* radiator loads in the same panel for comparison; the actual composite for each technology shows only its own waste-heat trace. **Physical meaning.** At the approximately constant 1-au orbit and with assumed perfect Sun tracking, the incident curve stays near 13.61 MW while generated PV output remains near 3.67 MW. The optical electronics/transmitter radiator rejects approximately 1.84 MW; the analogous microwave baseline has lower radiator waste heat because the configured electrical-to-beam conversion efficiency is higher. The difference between incident and PV output is *not wholly radiator heat*: some sunlight is reflected and PV-panel heat radiates from the PV plate itself.

What to check. Ensure that multiplying solar flux by 10,000 m^2 reproduces the roughly 13.61 MW incident value at 1 au. The PV power must account for absorptivity, conversion efficiency and temperature. The energy ledger must account separately for reflected light, PV heat and transmitter heat, without double-counting.

4.2 Panel M4-B: PV-temperature threshold over time

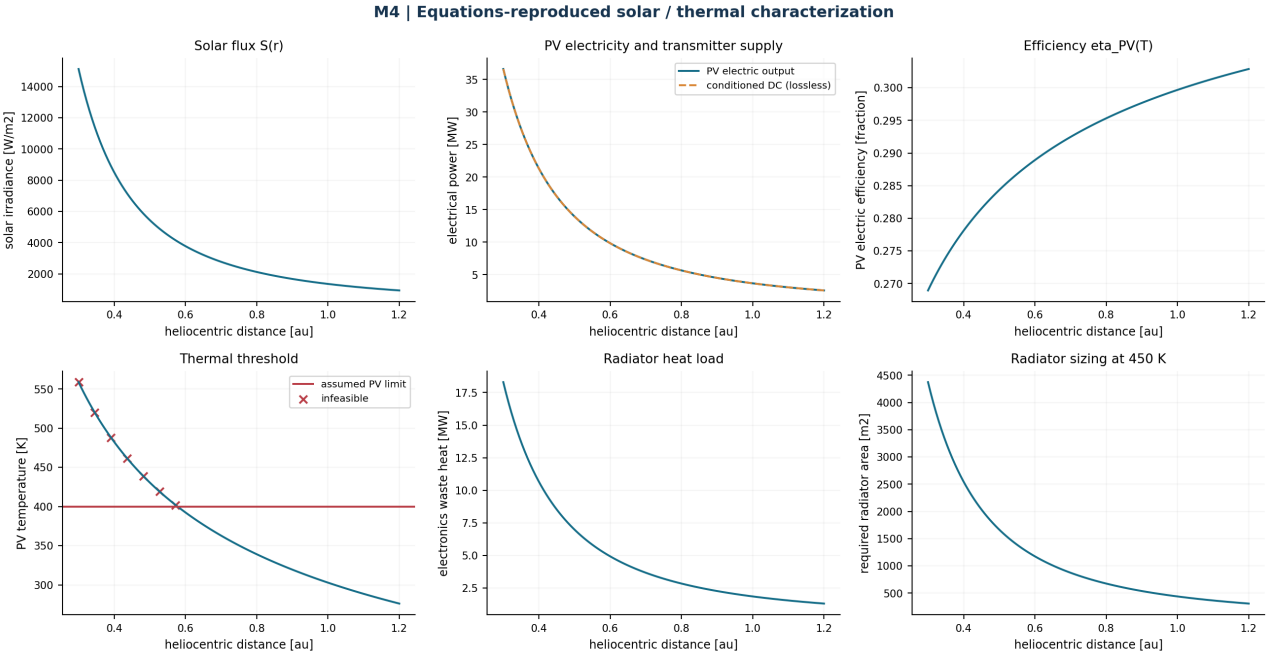
Horizontal axis: time since start [days] **Vertical axis**: PV temperature [kelvin, K]

The **blue line** is the equilibrium temperature of the modeled isothermal PV sheet, about 302.8 K. The **red line** is the

assumed maximum permitted PV temperature, 400 K (≈ 127°C). The figure title in the original optical composite additionally reports a **maximum required radiator area** about 438.5 m^2 and a **thermal-feasible fraction** of 1.000. In the microwave composite, that area is about 263.1 m^2. **Physical meaning.** The baseline thermal conditions remain below the configured cell limit for every epoch: the M4 model does not stop this 1-au design from operating. The radiator-area figure represents the area needed at the chosen 450 K radiator temperature limit, with ideal exposure to space; it is not a measured deployed mass.

What a successful result looks like

For M4, success means the thermal-balance root converges, energy is conserved and feasibility flags agree with temperature margins. It does *not* require the design to be feasible at every radius; correctly rejecting overheated configurations is a successful result.



4.3 Panel M4-C: Solar flux versus heliocentric distance

Horizontal axis: heliocentric radius r [au], about 0.3–1.2 **Vertical axis**: solar irradiance $S(r)$ [W/m^2]

The steep decline follows $S(r) = 1361(1\,au/r)^2$ W/m^2. **Solar irradiance** is incoming sunlight power per unit area *normal to the ray.* At 0.3 au it is around 15.1 kW/m^2, versus 1.36 kW/m^2 at 1 au. The curve is purely about incident solar flux, not the fraction reaching Earth. **Success**: correct inverse-square scaling and correct dimensional units.

4.4 Panel M4-D: Collector electrical power

Horizontal axis: heliocentric radius [au] **Vertical axis**: power available in the PV/electrical chain [MW]

The **solid blue** P_{PV} is electricity generated by the solar cells. The **dashed orange** $P_{dc\to tx}$ is that electricity after conditioning losses, before conversion into a beam. They overlap because the example's **power-conditioning efficiency is 1.0**, explicitly optimistic. Their steep decrease with distance mostly follows solar flux, partly offset by the PV efficiency's temperature dependence. **Physical meaning.** The highest plotted values at small r are *formal operating-point calculations*; they are not automatically usable because the PV temperatures there exceed a configured safety/qualification limit. The transmitter must be disabled when the thermal model declares the state infeasible.

4.5 Panel M4-E: Temperature-dependent PV conversion efficiency

Horizontal axis: heliocentric radius [au] **Vertical axis**: electricity divided by *absorbed* sunlight [dimensionless fraction]

The efficiency rises slightly with radius: lower solar heating yields lower PV temperature and better conversion under the assumed negative temperature coefficient. The code uses $\eta_{\mathrm{PV}}(T) = \mathrm{clip}[\eta_{\mathrm{ref}}(1 + \gamma_T(T - T_{\mathrm{ref}})), \eta_{\mathrm{min}}, \eta_{\mathrm{max}}]$. In the baseline $\eta_{\mathrm{ref}} = 0.30$, $T_{\mathrm{ref}} = 300$ K and $\gamma_T = -0.0004\ \mathrm{K}^{-1}$. **Clip** means the model limits its output to a prescribed allowed interval. **Physical meaning**. An efficiency near 0.30 means roughly 30% of *absorbed* solar power becomes electrical power. It must not be multiplied again as if it had been defined relative to already absorbed energy, and must not be confused with final grid efficiency.

4.6 Panel M4-F: PV thermal limit and red infeasible marks

Horizontal axis: heliocentric radius [au] **Vertical axis**: PV equilibrium temperature [K]

The blue curve falls as distance increases because radiative equilibrium responds approximately as $S^{1/4} \propto r^{-1/2}$ when conversion efficiency is slowly varying. The horizontal red line is the assumed 400 K PV limit; red x marks identify points above this limit. **Physical meaning**. The original recorded sweep gives approximately 559 K at 0.3 au, 431 K at 0.5 au, 363 K at 0.7 au, 303 K at 1.0 au, and 276 K at 1.2 au. Thus the 0.3 and 0.5 au configurations are rejected despite high theoretical generated power.

Assumptions / limitations

The 400 K threshold is a *chosen scenario parameter*, not a sourced maximum rating for a specific space-qualified PV module. Different materials, concentrators, heat transport and orientation laws may change the feasible range.

4.7 Panel M4-G: Radiator waste heat

Horizontal axis: heliocentric radius [au] **Vertical axis**: on-board electronics/transmitter waste heat [MW]

Waste heat grows nearer the Sun because more PV electricity is processed through conversion stages of non-unit efficiency. In the plotted optical scenario the main contribution is electrical-to-beam inefficiency of 50%, so roughly half of the conditioned DC becomes transmitter/radiator heat. The PV plate's own radiated heat is excluded from this curve; the two-node energy ledger accounts for it separately. **What to check**. When conversion efficiency is reduced, radiator heat must increase for the same PV output. Heat and outgoing aperture power plus losses must close the full energy budget.

4.8 Panel M4-H: Required radiator area

Horizontal axis: heliocentric radius [au] **Vertical axis**: required physical radiator surface area [m^2]

For emissivity ϵ, effective radiating sides n_f, view factor F and design limit T_{lim}, the required area is $A_{\mathrm{req}} = Q_{\mathrm{waste}}/[\epsilon\sigma n_f F(T_{\mathrm{lim}}^4 - T_{\mathrm{space}}^4)]$. Here σ is the Stefan–Boltzmann constant, T_{space} is background temperature and the radiator is assumed to radiate from both faces. **Emissivity** measures radiative effectiveness (0–1); **view factor** represents unobstructed radiative exposure. **Physical meaning**. A much larger heat load near the Sun demands thousands of square metres of radiator surface. At 1 au, the original optical result is about 438.5 m^2. An area curve cannot by itself tell whether the radiator's structural support, plumbing, mass and orientation are practical.

What a successful result looks like

The scientifically useful O2 output is not simply the highest PV-power curve. It is the *intersection* of strong generation, PV-temperature feasibility, radiator capacity and defensible deployed mass. The implemented M4 demonstrates a viable *model operating point* at 1 au, not a flight-qualified thermal system.

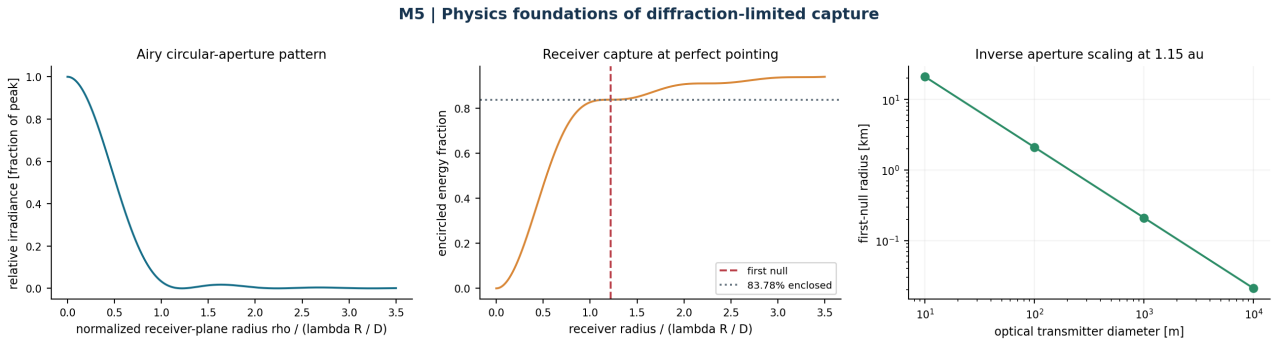
Research objective. The six-panel characterization answers O2 for the chosen idealized PV technology and energy-conversion efficiencies. It does not determine the best orbit once O3's transmission distance and O4's mass costs are included.

M5 – Diffraction, finite-receiver capture, pointing and wavefront errors

Key interpretation

Connection: M5 → O3. M5 asks what portion of the electromagnetic power leaving the spacecraft aperture lands on the receiver. M5 provides the fundamental spatial-loss factor later used by M6. In the source baseline, a poor optical or microwave capture fraction can dominate all other losses.

Figure status: Original M5 results appear in the bottom/middle portions of both committed composites and in two standalone M6b characterizations: pointing response and wavefront response.



5.1 Panel M5-A: Airy intensity profile of a circular transmitter

Horizontal axis: normalized distance $\rho/(\lambda R/D)$ from the beam center [dimensionless] **Vertical axis**: local intensity divided by peak intensity [dimensionless]

ρ is radial displacement from the beam axis on the receiver plane (m), λ transmitted wavelength (m), R propagation distance (m), and D transmitter aperture diameter (m). The ratio $\lambda R/D$ is the fundamental diffraction length. A **uniform circular aperture** produces the Airy pattern $I(\rho)/I(0) = [2J_1(x)/x]^2$, where $x = \pi D\rho/(\lambda R)$ and J_1 is a first-kind Bessel function of order one. **Physical meaning**. The central bright lobe contains most of the power, with weaker concentric rings. The intensity is maximum on axis and first vanishes at $\rho \approx 1.21967\lambda R/D$. The beam does *not* end at that radius: roughly 16.2% of the power lies outside the first dark ring.

What a successful result looks like

M5's Airy field must reproduce analytical Bessel-function roots, conserve total radiated power when integrated across the receiver plane, and give nonnegative intensity. A broad footprint is expected physics at large R, not a numerical failure.

5.2 Panel M5-B: Encircled energy / finite receiver capture

Horizontal axis: receiver radius expressed in units of $\lambda R/D$ **Vertical axis**: encircled beam-energy fraction [0–1]

Encircled energy $EE(a)$ is the proportion of transmitted power inside a centered circular disk of radius a. In the ideal Airy model, $EE(x) = 1 - J_0(x)^2 - J_1(x)^2$. At $x = 3.8317$ (first Airy dark ring), $EE \approx 0.8378$. For a tiny receiver inside a much wider spot, $EE \approx x^2/4$. **Physical meaning**. A larger receiver intercepts more energy until capture approaches

1. The receiver need not intercept 100% of the light simply because its radius is slightly larger than the first dark ring. This panel is the mathematically simplest link between transmitter diameter, range and receiver size.

What to check. Capture must be between 0 and 1; it must approach zero for a vanishing receiver and approach one as a sufficiently large receiver encloses the propagated pattern.

5.3 Panel M5-C: Inverse scaling with aperture diameter

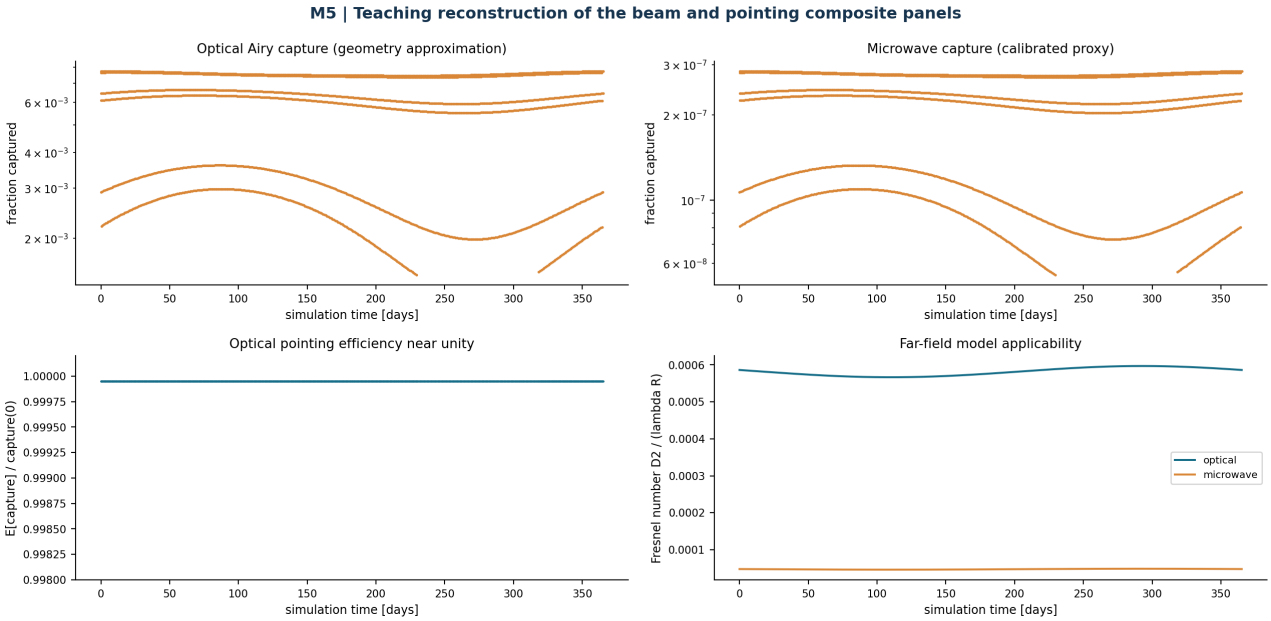
Horizontal axis: transmitter aperture diameter [m], logarithmic **Vertical axis**: first-dark-ring radius at the receiver [km], logarithmic

The equation $r_{\text{null}} = 1.21967\lambda R/D$ produces a decreasing straight line of slope −1 on log–log axes when λ and R are fixed. A tenfold diameter increase reduces the ideal first-null radius tenfold. This panel uses 1-μm light and an example 1.15-au range. It is not an engineering mass comparison: a 100-m deployable diffraction-limited optical aperture may be very difficult to manufacture and control.

Assumptions / limitations

The Airy approximation assumes uniform circular illumination and a scalar electromagnetic field in the far field. Real optics can have edge taper, obscuration, surface errors, thermal distortion, finite linewidth, partial coherence or pointing bias. The implementation includes several alternatives but not all hardware effects.

Figure status: The next four curves re-create the plotted variables of the two composite figures. They use simplified geometry and calibration; the original source images contain the authoritative time-sampled results.



5.4 Panel M5-D: Optical beam capture over time

Horizontal axis: days since simulation start **Vertical axis**: capture fraction η_{capture}, logarithmic

The **original optical composite's** bottom-left beam panel contains three features: a point series for *perfect pointing*, a point series for *expected (applied) capture*, and a shaded *p05–p95 band* for stochastic pointing error. The modeled transmitter diameter is 10 m, wavelength 1 μm, and receiver radius 1 km. The source report gives a median capture *during available epochs* of approximately 0.00631 (0.631%). **Physical meaning**. Even without gross pointing error, a 1-km-radius receiver intersects only a small portion of a diffraction-limited optical spot at ∼ 1.15 au. Different strands at a given season arise from daily changes in elevation and projected receiver area. Capture changes with geometry, not because the spacecraft is changing its PV capacity.

What to check. A successful plot shows zero or undefined capture during blocked epochs, finite fractions at accessible epochs, correctly normalized Airy energy and agreement of the deterministic jitter integration with cross-checks. A very small capture fraction is a valid negative feasibility result.

5.5 Panel M5-E: Microwave beam capture over time

Horizontal axis: elapsed days **Vertical axis**: receiver capture fraction, logarithmic

The **original microwave composite** uses a different system: 2.45-GHz wavelength $\lambda = 0.12236$ m, 1-km aperture with a tapered field, 10-km-radius horizontal rectenna, 70% electrical-to-RF conversion, and a specific sparse-lattice approximation. It reports a median captured fraction of about 3.88×10^{-7} during available epochs, or 0.0000388%. **Physical meaning.** The microwave wavelength is roughly 10^5 times the optical wavelength. Although the illustrative RF aperture is much larger, its beam diverges enough that almost all power misses the 10-km rectenna. The smaller y values are physically meaningful; the two baseline technologies are *not* compared at equal transmitter mass, aperture area or receiver infrastructure.

Assumptions / limitations

The teaching redraw uses a simpler calibrated envelope than the implemented tapered-aperture/sparse-array field. The exact microwave curves, their Fresnel numbers and their source annotations remain in the linked original microwave composite.

5.6 Panel M5-F: Pointing efficiency versus time

Horizontal axis: elapsed days **Vertical axis**: expected capture divided by perfect-pointing capture [dimensionless]

Pointing efficiency is $\eta_p = \mathbb{E}[\eta_{\text{capture}}]/\eta_{\text{capture,nominal}}$. In the original composite this curve is plotted with a *Matplotlib offset scale*, e.g. an optical annotation resembling $10^{-6} + 0.9995$. A displayed vertical tick should be interpreted as a small *offset* from that baseline, not a raw value of 7 or 8. The optical pointing efficiency is therefore very near 1 with the configured 10^{-9} rad per-axis jitter. **Physical meaning.** The baseline points near unity indicate little additional jitter loss *under the very precise assumed pointing control.* This does not demonstrate such control has been achieved in actual interplanetary hardware. Also, this ratio can exceed 1 for some ring-pattern and offset geometries: jitter need not always decrease energy intercepted by a disk that straddles an Airy ring.

5.7 Panel M5-G: Fresnel number

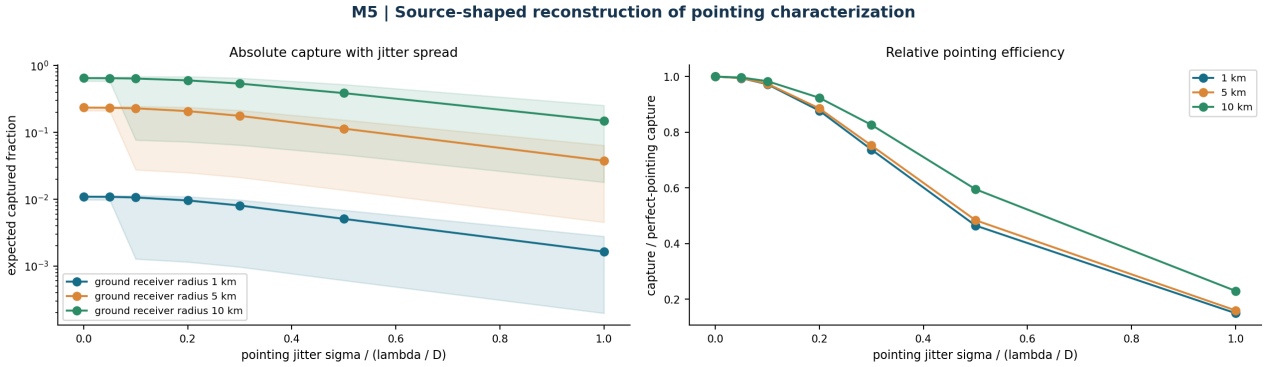
Horizontal axis: elapsed days **Vertical axis**: Fresnel number $N_F = D^2/(\lambda R)$ [dimensionless]

Fresnel number helps distinguish near-field and far-field electromagnetic propagation. In the originals, the Fresnel number is a *gray curve on the right vertical axis* of the pointing panel. The values for the modeled cases are small (order 10^{-4} to 10^{-5}), which supports use of the Fraunhofer far-field approximation. The software's far-field validity threshold is $N_F \leqslant 0.5$. **Physical meaning.** Changing $R(t)$ gently changes the Fresnel number. The RF and optical values are different because their wavelengths and apertures differ. A large Fresnel number would not simply imply poor capture; it would imply the current far-field model should be replaced with a near-field propagator.

What a successful result looks like

M5 success requires enforcing the model validity region instead of quietly using Fraunhofer equations where they no longer apply. The implementation provides a Fresnel–Hankel alternative and rejects invalid far-field configurations.

Figure status: The following two panels mirror the standalone `capture_vs_pointing.png` figure; lines are teaching approximations calibrated to the visible graph, not the source numerical rows.



5.8 Panel M5-H: Absolute capture versus pointing jitter

Horizontal axis: normalized pointing sigma $\sigma_\theta/(\lambda/D)$ [dimensionless] **Vertical axis**: expected capture fraction [dimensionless; logarithmic]

σ_θ is the standard deviation of the assumed independent Gaussian angular pointing errors in each of two orthogonal axes (radians). The denominator λ/D measures the beam's characteristic diffraction angle. Thus x = 1 indicates one-sigma pointing jitter comparable to the diffraction angle. The blue, orange and green curves represent ground receiver radii 1, 5 and 10 km, all for a *fixed* 1-au range and a zenith (90-degree elevation) receiver. **Physical meaning.** A larger receiving disk intercepts more beam power, producing a higher absolute capture curve. As jitter grows, the expected capture generally decreases for these centered, central-lobe scenarios. The shaded p05–p95 region summarizes *capture fluctuations under the specified pointing distribution*, not numerical or epistemic uncertainty in the full spacecraft model.

What to check. An ideal zero-jitter baseline must coincide with nominal capture; a centered receiver with a smooth central lobe should generally lose capture as jitter grows. Results must be stable under quadrature refinement and agree with analytic Gaussian-beam and sampled Monte Carlo checks where available.

5.9 Panel M5-I: Relative pointing efficiency versus jitter

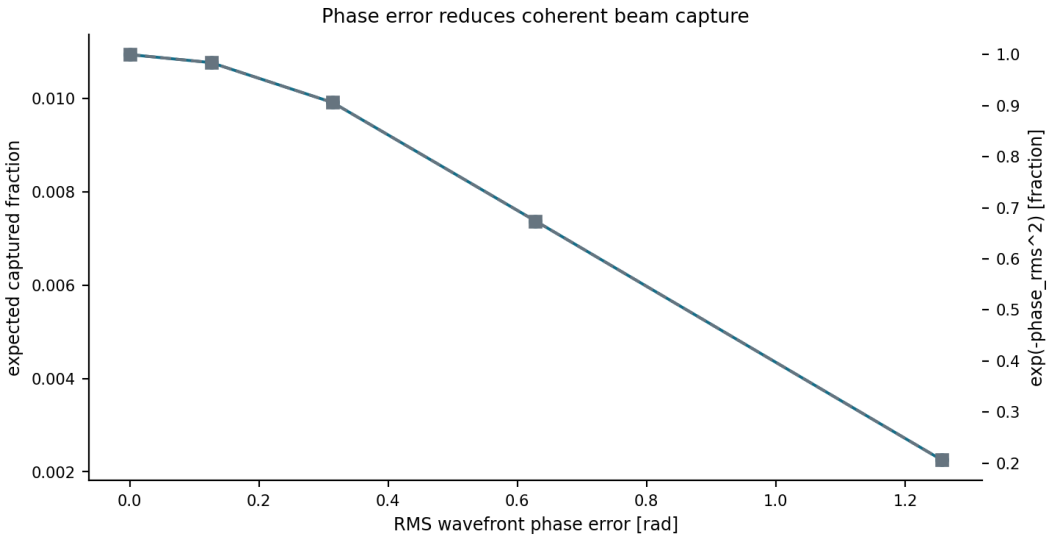
Horizontal axis: the same $\sigma_\theta/(\lambda/D)$ **Vertical axis**: ratio $\mathbb{E}[\eta_{\text{capture}}]/\eta_{\text{capture}}(0)$ [dimensionless, generally near 0–1]

The original right-hand plot divides each expected capture by that *same receiver size's own perfect-pointing capture*. Thus the curves all start near 1 even though their absolute capture values differ dramatically. At x near 1, the visible source graph shows roughly 0.15–0.23 remaining efficiency depending on receiver size. **Physical meaning**. If the 10-km receiver has 0.23 relative pointing efficiency, it does *not* mean the system captures 23% of the entire transmitted beam. It means its captured fraction is 23% of what that same receiver would capture at perfect pointing. The distinction between absolute and relative metrics is essential when comparing designs.

Assumptions / limitations

The M6b figure is a *sensitivity plot at one explicit fixed link geometry*. It is not a year-long ground-availability calculation and does not account for clouds or the changing real trajectory.

Figure status: The following graph reproduces the implemented *functional law* for wavefront loss, using the plotted phase-error values. Original: capture vs wavefront.



5.10 Panel M5-J: Expected capture versus RMS wavefront error

Horizontal axis: RMS wavefront phase error $\phi_{\mathrm{rms}} = 2\pi\sigma_w/\lambda$ [radians] **Vertical axis**: left: expected capture fraction; right: Ruze/Strehl factor [dimensionless]

RMS means root mean square. σ_w is the RMS optical path-length deviation from an ideal wavefront (m), not the reflecting surface displacement. The **Strehl ratio** approximates the central coherent peak intensity retained in the presence of small random wavefront error. The source uses $\eta_w = \exp[-(2\pi\sigma_w/\lambda)^2] = \exp(-\phi_{\mathrm{rms}}^2)$ and applies this factor to the captured main-beam power. **Physical meaning**. The **left y axis** shows the actual expected intercepted beam fraction, falling from about 1.1% to around 0.23% in the plotted example. The **right y axis** shows the fractional main-lobe retention, falling from 1 to approximately 0.21 when $\phi_{\mathrm{rms}} \approx 1.26$ rad. The two curves share one x axis but have different scales. Do not compare their raw numerical heights.

What to check. At zero wavefront error the factor must be 1. At $\phi_{\mathrm{rms}} = 1$, it should equal $e^{-1} \approx 0.3679$. Capture should change consistently with the source's single-loss-factor model.

Assumptions / limitations

The model treats wavefront-aberrated power as scattered outside the useful receiver; it does not propagate an arbitrary complex distorted wavefront or model thermal blooming, adaptive correction, polarization and the entire sidelobe field. The source labels the scalar diffraction calculations as idealized upper bounds on certain peak-intensity quantities.

5.11 How to interpret atmospheric attenuation across M5 panels

The atmosphere is *not* part of the definition of η_{capture}; it multiplies power separately in M6. In a simple scenario model, optical zenith transmittance τ_z is converted to an elevation-dependent factor $\tau_z^{1/\sin e_l}$. In the RF model, attenuation is often specified in **decibels**: a loss A dB corresponds to a power multiplier $10^{-A/10}$. A 10-dB loss means only 10% of incident power remains; 3 dB is approximately one-half. **Physical meaning**. For optical transmission, a permanently `clear` weather setting is an optimistic service assumption. A model could show excellent diffraction capture but zero delivery on a cloudy day; these are separate factors in a sound link budget. The microwave option applies an ITU-R long-term attenuation exceedance setting, not a stochastic hour-by-hour rain/cloud process.

What a successful result looks like

For O3, the scientifically meaningful result is a map of absolute captured beam power and its sensitivity to distance, wavelength, aperture, pointing, wavefront and receiver size. A functioning M5 beam solver is demonstrated by analytical and independent optical checks; actual high-power beaming feasibility remains unproven.

Research objective. M5 substantially addresses the electromagnetic-capture component of O3. It does not independently determine final delivered grid watts (M6), transmitter mass (O4), realistic weather availability, or comparison to

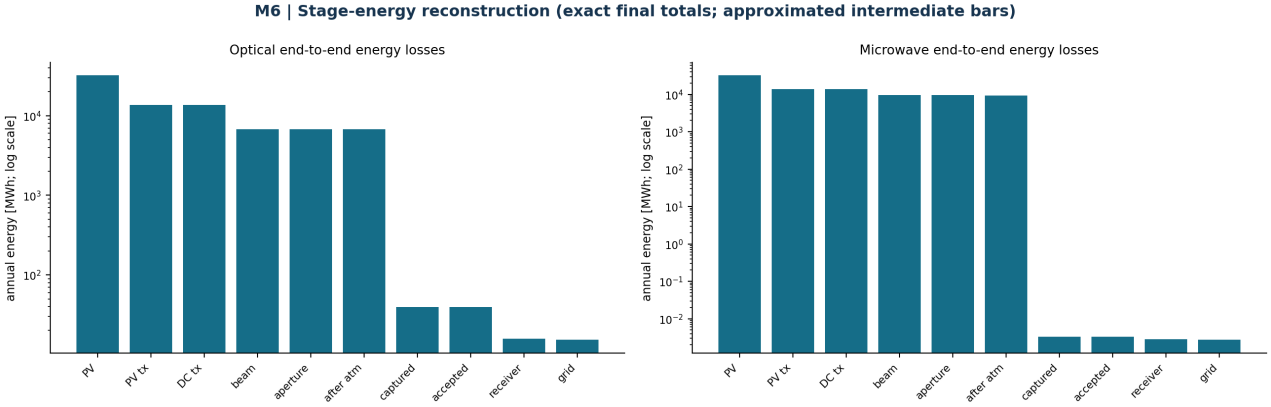
GEO (O5).

M6 – Reading the complete link budget, annual grid power and sweep outputs

Key interpretation

Connection: M6 → O3, with preliminary O4 evidence. M6 integrates orbit, access, PV conversion, radiators, diffraction, atmosphere and receiver conversion into $P_{\mathrm{grid}}(t)$. It can answer what one *specified* heliocentric spacecraft would deliver under the chosen assumptions. It cannot by itself answer multi-satellite scheduling (M7), constrained architecture optimization (M8) or a GEO comparison (O5).

Figure status: The original M6 composite graphs are optical and microwave. The following stage charts preserve the recorded initial/final totals, while intermediate bars are approximate reconstructions rather than source-extracted time-series sums.



6.1 Panel M6-A: The optical power-stage "waterfall"

Horizontal axis: ordered stages of the electricity/beam/receiver chain **Vertical axis**: annual energy through each stage [MWh; logarithmic]

The source calls this a **waterfall**, but each bar represents *energy still remaining* at a given stage, not the incremental loss at that stage. The ten labels are: **PV** (electricity generated); **PV tx** (PV electric energy when transmitter is enabled); **DC tx** (conditioned electricity); **beam** (electromagnetic power before aperture losses); **aperture** (power emitted); **after atm** (after atmospheric attenuation); **captured** (power physically intercepted by the receiving area); **accepted** (power not clipped by receiver limits); **receiver** (converted into usable electric power); and **grid** (after grid conditioning). **Physical meaning**. There are three conceptually separate losses. *Availability* causes the first major decrease from year-round generated PV energy to electricity allocated to active transmission. *Conversion* causes a further drop when only 50% of conditioned DC becomes an optical beam. *Diffraction and finite capture* cause the decisive drop between **after atm** and **captured**: most beam energy misses the 1-km-radius receiver. Receiver conversion (40%) and grid conditioning (97%) reduce the captured fraction further.

What to check. Read the logarithmic y axis carefully: dropping from 10^4 to 10^1 MWh is a thousandfold decrease. Every passive or lossy stage must satisfy $E_{k+1} \leqslant E_k$ in the numerical implementation. The original optical final bar is approximately **15.35 MWh** over the year.

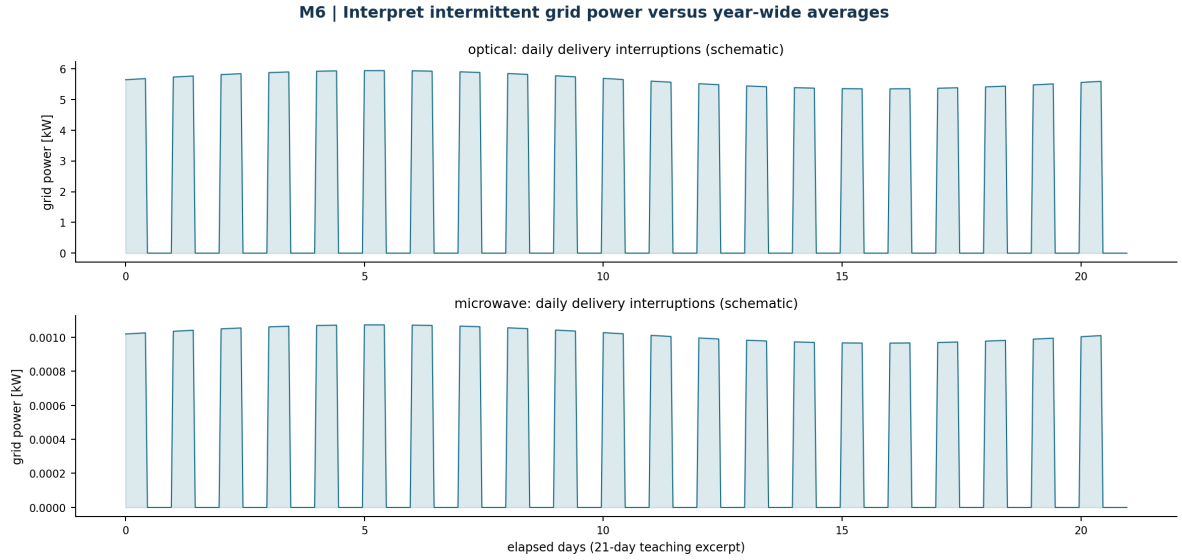
6.2 Panel M6-B: Microwave power-stage waterfall

Horizontal axis: the same ten sequential power stages **Vertical axis**: energy [MWh; logarithmic]

The microwave system has a different transmitter conversion efficiency (70%), ground receiver (rectenna) conversion efficiency (85%), RF aperture, and atmospheric attenuation model. Despite higher selected conversion efficiencies, its beam-capture fraction is extremely small at interplanetary range. The final energy is about **0.002757 MWh**, or 2.757 kWh, over the year. **Physical meaning.** The RF waterfall highlights why improving electrical-to-beam conversion alone cannot rescue a design when finite receiver interception is many orders of magnitude below unity. A few stages can be very efficient and the whole chain still deliver almost nothing.

What a successful result looks like

M6 success requires strict energy bookkeeping, the same saved-power data feeding both bars and summary metrics, correctly applied receiver caps, zero downstream power when unavailable, and no double counting of optical aperture, wavefront, lattice, atmosphere or pointing losses.



6.3 Panel M6-C: Optical grid-power time series

Horizontal axis: time since simulation start [days] **Vertical axis**: electricity reaching the grid P_{grid} [kW]

Grid power is an instantaneous rate, not energy. A value of 5 kW at a particular epoch means energy is arriving at 5 kJ/s at that instant. The original composite spans the full year and plots all sampled grid-power values. Dense hourly oscillations make the trace look like a solid block with a varying upper boundary, but the satellite has frequent *zero-power outages*. The zoomed illustration here depicts this duty cycling; its exact pass timings are *not original data*. **Physical meaning.** The recorded optical baseline has mean grid power 1751 W (1.751 kW), peak approximately 5937 W, *median zero*, and availability about 0.4221. Its 95th percentile is about 5647 W. The mean is far below the typical nonzero operating value because delivery is repeatedly interrupted.

What to check. The correct time average is the area under the $P_{\text{grid}}(t)$ curve divided by the elapsed time, including all zero-power samples. It is *not* the average of the top boundary or of only the nonzero points. With uniform sampling step Δt, the code uses endpoint half-weights to compute $E = \sum_k w_k P_k$ and $\overline{P} = E/T$.

6.4 Panel M6-D: Microwave grid-power time series

Horizontal axis: elapsed days **Vertical axis**: grid power [kW]; note that 0.001 kW = 1 W

The microwave trace has the *same access timing* as the optical case because the baseline geometry and ground site are shared, but much smaller nonzero power. The documented mean is approximately **0.3145 W**, peak about 1.073 W and median zero. The line height is small: an ordinate labeled 0.0010 kW is 1 W, not 1 kW. A scale change between technology plots can visually disguise the enormous gap in delivered watts. **Physical meaning.** The microwave result is a *negative*

baseline feasibility result for the specific 1-km, 2.45-GHz transmitter and 10-km rectenna at about 1.15 au, not a proof that all RF concepts are impossible.

6.5 Panel M6-E: Numerical summary text box in both composites

Recorded source metric	Optical baseline	Microwave baseline
Mean grid power	1751 W	0.3145 W
Median grid power	0 W	0 W
Peak instantaneous grid power	5937 W	1.073 W
Availability	0.4221	0.4221
Delivered electrical energy (year)	15.35 MWh	0.002757 MWh
End-to-end energy efficiency	1.29×10^{-4}	2.31×10^{-8}
Longest outage	14 h	14 h
Number of reported outages	366	366

Mean is energy divided by elapsed time; **median** is the 50th percentile; **peak** is the maximum sampled power; **p95** is the 95th percentile. **Availability** is the fraction of time with no configured blocking reason. **End-to-end energy efficiency** is total grid energy divided by the total sunlight incident on the collector over the window, not PV efficiency or beam interception fraction. **Physical meaning.** The 3.67 MW mean PV electricity is a *generation* metric; comparing it with 1.751 kW mean optical grid output reveals very poor downstream delivery. Neither power figure by itself is a validated mass or economic metric.

6.6 Panel M6-F: Capacity factor in the grid-power plot title

The original optical plot title reports a capacity factor near 0.297 and the microwave counterpart about 0.295. **Capacity factor** is $CF = \overline{P}_{\mathrm{grid}}/P_{\mathrm{rated}}$. Here *rated* is a **reference link** evaluated at mean Sun distance and mean range, a zenith-facing receiver, full illumination, configured conversion factors and relevant receiver limits. It is *not* simply a manufacturer's PV nameplate rating, nor 3.67 MW. Therefore a capacity factor around 0.3 must never be interpreted as 30% sunlight-to-grid efficiency.

6.7 Panel M6-G: The composite's idealized-model notes

The bottom-right text panel explicitly identifies assumptions such as scalar diffraction without atmospheric turbulence and source-reported optimistic settings. The warning about the largest unsampled diurnal hour-angle gap (about 8.64° in the reported composites) is a **sampling diagnostic**; a small plotted Monte Carlo-like variation must not obscure an underresolved day/night pattern.

Assumptions / limitations

The baseline scenarios assume clear optical weather for the whole year, no receiver-side beam-density or heat-rejection limit, perfect Sun tracking, idealized transmitter optics, and no operational Sun-avoidance angle. These are *optimistic or missing constraints.* Excellent code validation does not turn them into realistic physical performance guarantees.

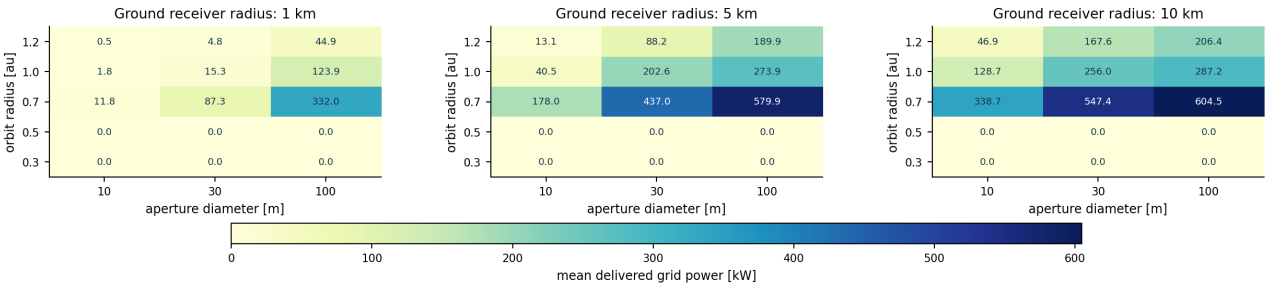
PLOT INTERPRETATION 7

M6 parameter sweeps – a bridge toward O4, not constellation optimization

Key interpretation

A **parameter sweep** evaluates many independent configurations on a discrete grid. An **optimizer** solves for best trade-offs, often subject to mass, receiver, pointing, safety or reliability constraints. A **constellation** contains many satellites sharing orbital geometry and receivers. The current 45+15 point sweeps are single-satellite evidence; they are not M7/M8 multi-satellite optimization.

Figure status: Unlike the schematic time histories, these heatmaps are re-plots of *recorded values* from the two committed CSV files. Source: 45 optical runs and 15 RF runs.



7.1 Panel M6-H: Optical mean grid-power heatmaps

Horizontal axis: transmitter aperture diameter [m], at tested values 10, 30, 100 **Vertical axis**: heliocentric orbital radius [au], tested at 0.3, 0.5, 0.7, 1.0 and 1.2

There are **three heatmaps**: left for a 1-km-radius receiver, center for 5 km and right for 10 km. **Color** is the same quantity across all three: mean delivered grid power [kW] over the year. Each numeric cell states the recorded power, in kW. Darker/lighter regions correspond to the color-bar scale; exact numbers are more reliable than qualitative color intensity. **Physical meaning**. At 0.3 and 0.5 au, every optical cell is zero because the PV temperature violates its assumed limit and the transmitter is disabled. At 0.7 au, larger transmitter diameter and receiver radius improve capture strongly; the highest power among the *tested* points is 604.520 kW for the 100-m optical transmitter and 10-km receiver. At 1.0 au, that same geometry delivers about 287.234 kW.

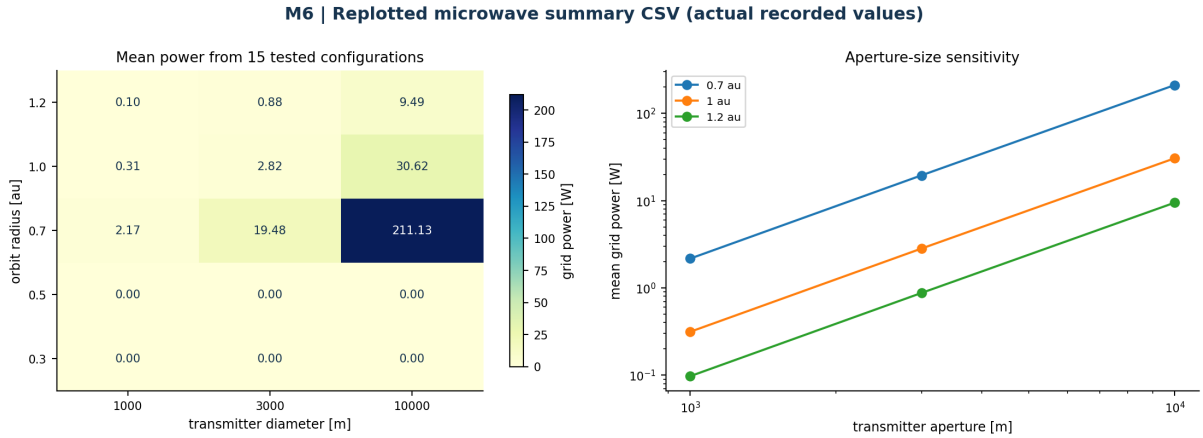
Why these axes? Increasing D shrinks the diffraction angular scale λ/D. Increasing receiver radius a captures a larger area of the spot. Moving radially changes both intercepted solar flux $\propto r^{-2}$ and the Earth-link geometry. Therefore delivered power need not be monotonic in orbit radius, even if solar flux is.

What a successful result looks like

A credible sweep must be reproducible, include all hard feasibility flags, use comparable assumptions and show the actual optimum only *within the searched grid.* A best point should survive realistic Sun-avoidance, weather, receiver area, thermal and mass limits. The current sweep does *not* show the latter.

Assumptions / limitations

The 0.7-au optical winner uses a 100-m laser aperture and a 10-km-radius ground receiver. The repo notes problematic near-Sun/solar-transit geometry because `sun_avoidance_deg=0`. It is not evidence of a feasible, authorized, continuously operating 604-kW plant. The receiver area is enormous (about $\pi(10\,\mathrm{km})^2 \approx 314\,\mathrm{km}^2$), before practical infrastructure and cost.



7.2 Panel M6-I: Microwave sweep heatmap

Horizontal axis: transmitter aperture diameter [m]: 1000, 3000 or 10000 **Vertical axis**: orbital radius [au]

Color bar: annual-average grid output [*watts*], not kW. Each cell contains a recorded result. The receiver remains fixed at a 10-km radius. Rows 0.3 and 0.5 au are zero for the same PV thermal-limit reason; 0.7, 1.0 and 1.2 au have nonzero values. **Physical meaning**. The largest tested mean is about **211.13 W** at 0.7 au with a 10-km transmitter aperture. At 1.0 au, that 10-km aperture yields about 30.62 W. The change when diameter increases from 1 km to 10 km is huge *multiplicatively* but tiny *absolutely* relative to MW-level generated PV electricity.

7.3 Panel M6-J: Microwave size-sensitivity line plot

Horizontal axis: aperture diameter [m], logarithmic **Vertical axis**: mean grid power [W], logarithmic

The three curves correspond to the 0.7, 1.0 and 1.2 au tested radii. The visible near-quadratic trend is consistent with a small-aperture-capture regime in which intercepted power depends approximately on D^2 for fixed wavelength, distance, and receiver area. A 10× diameter increase in such a regime could deliver roughly 100× capture efficiency; the actual observed ratios also depend on geometry and other losses. **Physical meaning**. A high relative improvement should never be mistaken for a practical absolute result. Even the largest plotted RF mean is in watts, not megawatts. Conversely, the plot cannot rule out different carrier bands, physically different receiver geometries or different deployment concepts without evaluating them.

7.4 What O4 needs beyond these sweeps

A truly successful O4 constellation study should produce figures of *grid power versus time for many satellites*, fraction of the year above a contracted MW-level target, receiver capacity/scheduling violations, collective longest outages, and a **Pareto frontier**: the set of designs for which no target metric can be improved without worsening another. Relevant trade-offs include annual grid energy, spacecraft and receiver area, total deployed mass, thermal feasibility, outage probabilities, pointing-control requirements and safety limits. Such figures are *not in the inspected branch* because M7 and M8 remain open. **Research objective**. These sweeps are preliminary O4 design-space exploration, but they only validate the *single-link model's response* to parameters. A research objective about a *Dyson swarm* must ultimately combine multiple satellites and multiple receivers; that code is not yet implemented.

Crosswalk: which plots answer which research objective?

Objective	Milestones	Best evidence currently present	Scientific coverage
O1 Orbital architecture	M1–M3	Kepler validation, REBOUND checks, orbit projection, Earth range, access/elevation, light time	Good *single-satellite* foundation; not optimal swarm
O2 Solar and thermal	M4	Six-panel radius characterization, baseline PV/radiator traces, thermal feasibility flags	Computed under assumed PV/radiator technology
O3 Interplanetary transmission	M5–M6	Optical/RF beam capture, pointing and wavefront responses, stage-energy losses, grid-power time series	Strong idealized single-link analysis; real hardware incomplete
O4 Constellation optimization	M6 preliminary; M7–M8 pending	Optical 45-point and microwave 15-point CSV sweeps	No multi-satellite scheduling or constrained Pareto optimization
O5 GEO comparative feasibility	M9/later	No matched GEO simulation or side-by-side performance plot	Unanswered

8.1 Do not confuse the three meanings of "successful"

(i) Valid computation: unit conversions, coordinate origin, conservation, numerical convergence, physical ranges, power monotonicity and reproducibility pass tests. **(ii) Informative scientific experiment**: the output isolates a useful constraint, including a negative finding such as "almost all RF beam energy misses the receiver." **(iii) Feasible engineering design**: the same result survives realistic materials, mass, power-density, weather, receiver, safety, regulatory and availability assumptions. The current repo is strongest at (i) and is producing evidence toward (ii), with (iii) still unresolved.

8.2 Practical checklist for a manuscript figure

1. **State what is plotted**: give the horizontal/vertical variables, SI units, frame of reference, time origin and whether the axis is logarithmic.
2. **Identify every series**: solar flux is not PV power; absolute beam capture is not relative pointing efficiency; availability is not capacity factor.
3. **Declare source and uncertainty**: is the curve a saved numerical run, an analytical validation, a fixed-geometry characterization, a derived CSV heatmap or only a teaching illustration?
4. **State the claim and its falsifier**: for example, "fixed 1-km receiver captures less than 1% of the ideal optical beam," subject to aperture and range assumptions.
5. **Check physical constraints**: thermal failure, Sun separation, Earth occultation, cloud cover, receiver saturation, heat rejection, allowed peak irradiance and pointing-control performance.
6. **Compare to a meaningful benchmark**: report annual delivered Wh, W per unit infrastructure, service availability

and ultimately an equivalent GEO reference.

PLOT INTERPRETATION A

Exact original plot and report catalogue

Repository branch: https://github.com/extremefattypunch/Project-Dyson-Sphere/tree/helioswarm-m0-m6. The pages cited below are online source locations, not embedded originals. Opening links may require authorization for the private repository.

Source item	Direct URL and purpose
Optical composite	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/figures/m6/mvp_optical_composite.png
Microwave composite	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/figures/m6/mvp_microwave_composite.png
Solar / thermal radius sweep	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/figures/m6b/power_thermal_vs_distance.png
Capture versus pointing jitter	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/figures/m6b/capture_vs_pointing.png
Capture versus wavefront error	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/figures/m6b/capture_vs_wavefront.png
Optical 45-point CSV	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/figures/m6/optical_1au_sweep_summary.csv
RF 15-point CSV	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/figures/m6b/microwave_1au_sweep_summary.csv
M1 implementation and tests	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/docs/milestones/M1.md
M2 implementation and tests	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/docs/milestones/M2.md
M3 implementation and tests	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/docs/milestones/M3.md
M4 implementation and tests	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/docs/milestones/M4.md
M5 implementation and tests	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/docs/milestones/M5.md
M6 implementation and tests	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/docs/milestones/M6.md
M6b figure characterization	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/docs/milestones/M6b.md
Physical assumptions	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/docs/model_assumptions.md
Validation report	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/docs/validation_report.md
Experiment log E1–E7	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/experiments/README.md
Plotting implementation	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/src/helioswarm/plotting.py
Characterization implementation	https://github.com/extremefattypunch/Project-Dyson-Sphere/blob/helioswarm-m0-m6/src/helioswarm/characterize.py

PLOT INTERPRETATION B

Terminology quick-reference

Term or symbol	Meaning in these figures
a (orbital)	Semimajor axis, ellipse size; [m] or [au].
e (orbital)	Eccentricity, shape parameter; dimensionless.
ν (true anomaly)	Physical orbital angle measured from perihelion; rad.
M, E	Mean and eccentric anomalies in Kepler's equation; rad.
$GM_\odot$	Sun's gravitational parameter; m^3/s^2.
J2000, ECLIPJ2000	Fixed inertial reference orientations; J2000 is not itself a physical origin.
SSB	Solar System Barycenter, the position origin for combined Cartesian states.
TDB, UTC	Dynamical uniform-like ephemeris time and civil coordinated time, respectively; must be converted explicitly.
β_{SRP}	SRP-to-gravity acceleration ratio, distinct from relativistic velocity v/c.
SRP	Solar radiation pressure, photons pushing on spacecraft surfaces.
Elevation	Receiver-local angle above horizon, degrees.
Occultation	Ray blocked by an intervening opaque body; affects link availability.
Light time	Beam vacuum travel delay between emission and ground arrival, seconds.
Aberration	Direction transformation between moving inertial frames caused by Lorentz kinematics.
$S(r)$	Solar irradiance [W/m^2] as a function of Sun distance.
P_{PV}	PV-generated electrical power [W], before transmission losses.
Q_{waste}	On-board electronics/transmitter waste heat [W] assigned to the radiator.
η	Efficiency fraction. Subscripts name each distinct physical stage.
Absorptivity α	Fraction of incident sunlight absorbed by an opaque PV sheet.
Emissivity ϵ	Thermal-radiation efficiency relative to an ideal black body.
View factor	Fractional geometrical radiative exposure to space.
D, λ, R	Transmitter diameter, carrier wavelength, and transmission range [m].
Airy pattern	Circular-aperture far-field intensity pattern with central lobe and rings.
Bessel J_0, J_1	Special functions of orders 0 and 1 governing circular-aperture diffraction.
Fraunhofer region	Far-field regime where transmitted amplitude is described by the aperture's Fourier transform.
Fresnel number	$D^2/(\lambda R)$, dimensionless propagation-regime measure.
Capture fraction	Fraction of beam power intercepted by the finite receiver [0–1].
Pointing sigma	One-axis standard deviation of Gaussian angular jitter [rad].
Strehl / Ruze factor	Assumed coherent-peak degradation from RMS wavefront phase error.
Rectenna	RF antenna array with rectifiers converting collected RF to DC.
Atmospheric attenuation	Energy lost or diverted along the atmospheric path, often expressed in dB.
Capacity factor	Annual mean grid power divided by reference-link rated grid power.
Availability	Weighted time fraction when no modeled hard constraint blocks link operation.
End-to-end efficiency	Year-delivered grid energy / incident collector sunlight energy.
Diurnal aliasing	Bias from sampling almost the same hour angles each solar day.
Pareto frontier	Set of non-dominated multiobjective designs; requires M8 optimization.
O1–O5	Scientific objectives, distinct from software milestones M1–M6.
M0–M9	Staged software development and research deliverables.
E1–E7	Recorded numerical experiment identifiers.
L0–L3	Orbital model-fidelity levels (two-body through refined perturbations).

Final synthesis: what the existing figures actually prove

Key interpretation

The clearest existing conclusion is that a heliocentric collector can generate MW-scale electricity in the specified model while delivering orders of magnitude less to a terrestrial grid because of geometry, duty cycle, conversion and diffraction. In the single-satellite optical baseline, the simulated collector produces about 3.67 MW PV electricity yet only 1.751 kW mean terrestrial grid power; the illustrative RF case yields about 0.3145 W. These results are valid findings *within the modeled assumptions*, not evidence of a commercially feasible system.

For a manuscript, the current figure set is strongest at demonstrating a reproducible *single-link feasibility model* (O1, O2, O3). The 45 optical and 15 microwave sweep summaries provide useful preliminary sensitivity evidence (O4) but no actual constellation optimization. There is no fair matched GEO benchmark yet (O5). The next scientific priority is to enforce sourced weather, Sun-avoidance and receiver limits, then implement M7 multi-satellite scheduling and M8 constrained optimization before asserting any net heliocentric advantage.